Regularization in Machine Learning

Lecture 1. Regularization Foundations

Sonnet 5 High

Claude Cowork

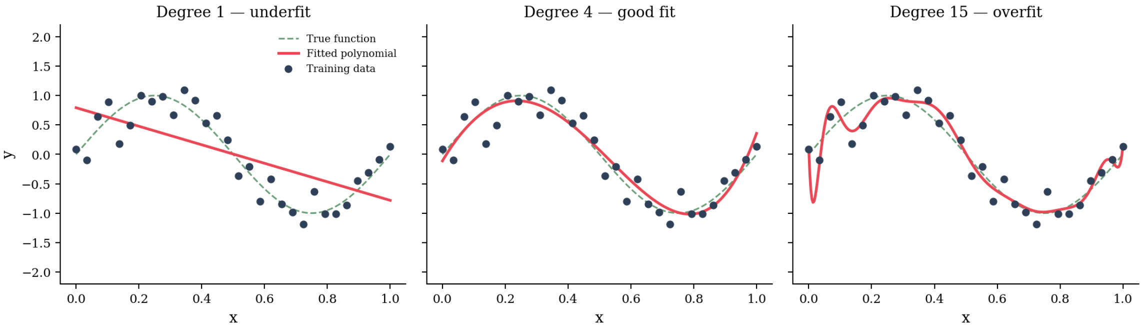
August 6, 2026

Outline

- Why models overfit & the bias–variance tradeoff
- Ridge and Lasso: mechanics and geometry
- Choosing λ and the Bayesian view
- Elastic net teaser & wrap-up

Why do models overfit?

More parameters give a model more capacity to fit the training data — including its noise, not just the underlying pattern.



The bias-variance tradeoff

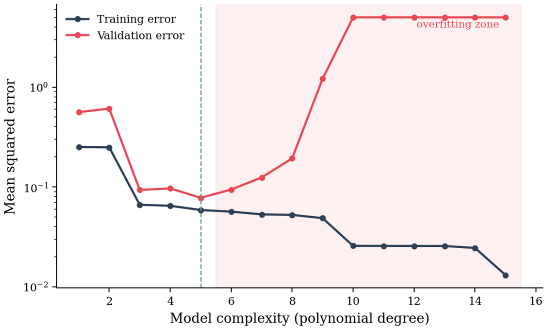
Expected test error decomposes into three additive terms:

$$\mathbb{E}\left[(y - \hat{f}(x))^2\right] = \underbrace{\mathrm{Bias}[\hat{f}(x)]^2}_{\text{model too simple}} + \underbrace{\mathrm{Var}[\hat{f}(x)]}_{\text{sensitive to training sample}} + \underbrace{\sigma^2}_{\text{irreducible noise}}$$

- **Bias:** systematic error from a model that can't capture the true pattern
- **Variance:** how much the fitted model changes across different training samples
- Regularization trades a small increase in bias for a larger reduction in variance

Overfitting in numbers

- Training error keeps falling as complexity grows
- Validation error is U-shaped — it eventually rises
- The growing gap between the two curves is the signature of overfitting



What is regularization?

Definition

Any technique that discourages a model from fitting the training data too closely, without changing the model family — typically by adding a penalty or constraint tied to model complexity.

- Alternative: manually pick a simpler model (fewer features) — crude and inflexible
- Regularization lets a tunable strength decide how much complexity the data actually supports

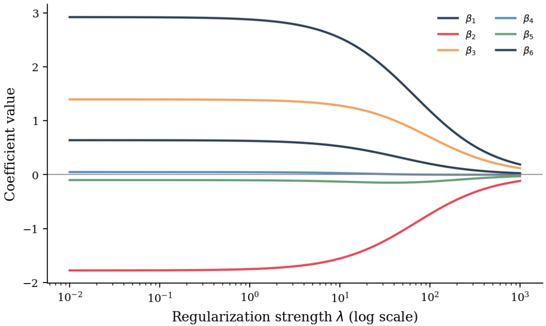
Ridge regression: the L2 penalty

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \underbrace{\|y - X\beta\|_2^2}_{\text{fit to data}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{penalty}}$$

- $\lambda = 0$: recovers ordinary least squares exactly
- $\lambda \to \infty$: all coefficients pushed toward zero
- The intercept is conventionally left out of the penalty

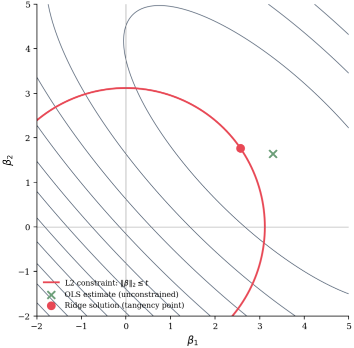
Ridge: effect on coefficients

- As λ grows, every coefficient shrinks smoothly toward zero
- Coefficients essentially never land exactly on zero
- Ridge keeps all features, just with smaller weight



Geometric picture: L2

- Ridge is exactly equivalent to minimizing the loss subject to $\|\beta\|_2 \leq t$
- The solution sits where the smallest loss contour touches the circular boundary
- A circle has no corners — the tangency point generically has both coordinates nonzero



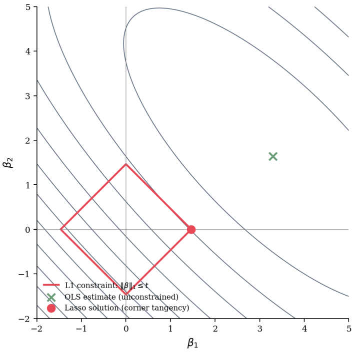
Lasso regression: the L1 penalty

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \underbrace{\|y - X\beta\|_2^2}_{\text{fit to data}} + \lambda \underbrace{\|\beta\|_1}_{\text{penalty}}$$

- Same idea as ridge — penalize the loss — with a different norm
- $\|\beta\|_1 = \sum_j |\beta_j|$, the sum of absolute values
- Unlike the L2 penalty, this norm can push coefficients to exactly zero

Geometric picture: L1 and sparsity

- Lasso is equivalent to minimizing the loss subject to $\|\beta\|_1 \leq t$ — a diamond-shaped region
- The loss contour typically first touches the diamond at a corner
- At a corner, one coordinate is exactly zero — lasso performs automatic feature selection



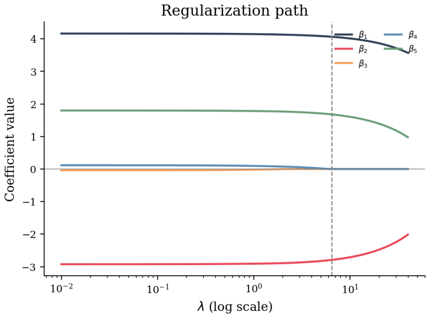
Ridge vs. Lasso

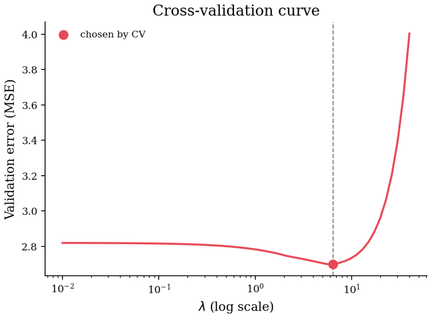
	Ridge (L2)	**Lasso (L1)**
Sparsity	No	Yes
Correlated features	Shrinks together	Picks one, zeros others
Solution	Always unique	May not be unique
Computation	Closed form	Iterative (e.g. coordinate descent)

Rule of thumb: use lasso when you suspect many irrelevant features; use ridge when most features carry some signal.

Choosing the regularization strength λ

λ is a hyperparameter — it is chosen by evaluating held-out performance across a grid of values via cross-validation, not learned from the training objective directly.





Regularization as prior belief

Key idea

Adding a penalty term to the loss is mathematically equivalent to placing a prior distribution over the parameters and computing the maximum a posteriori (MAP) estimate.

- “Penalty strength” becomes “how strongly we believe, before seeing data, that coefficients should be small”
- This reframes ridge and lasso as different *prior beliefs* about the coefficients

MAP interpretation: Ridge $\leftrightarrow$ Gaussian prior

Assume each coefficient has an independent prior $\beta_j \sim \mathcal{N}(0, \tau^2)$. The MAP estimate maximizes the posterior, equivalent to minimizing:

$$-\log p(y \mid X, \beta) - \log p(\beta) \;=\; \underbrace{\|y - X\beta\|_2^2}_{\text{data term}} + \underbrace{\frac{\sigma^2}{\tau^2}\|\beta\|_2^2}_{=\text{ ridge penalty}} + \text{const}$$

- The Gaussian prior is symmetric and “soft” around zero
- This is exactly why ridge shrinks smoothly rather than zeroing out coefficients

MAP interpretation: Lasso $\leftrightarrow$ Laplace prior

Assume each coefficient has an independent prior $\beta_j \sim \text{Laplace}(0, b)$. The MAP estimate is equivalent to minimizing:

$$-\log p(y \mid X, \beta) - \log p(\beta) \;=\; \underbrace{\|y - X\beta\|_2^2}_{\text{data term}} + \underbrace{\frac{2\sigma^2}{b}\|\beta\|_1}_{=\text{ lasso penalty}} + \text{const}$$

- The Laplace prior has a sharp peak at zero
- This is the probabilistic reason lasso favors exact zeros — the same conclusion as the geometric picture, from a different angle

Putting it together

Three equivalent ways to understand regularization:

- **Bias-variance:** controls model complexity to reduce variance
- **Geometric:** constrains coefficients to a norm ball (circle vs. diamond)
- **Bayesian:** encodes a prior belief that coefficients should be small (Gaussian vs. Laplace)

All three give the same answer — keep whichever lens is most intuitive.

Beyond L1/L2: elastic net

- Combining both penalties gets some of lasso's sparsity together with ridge's stability under correlated features
- Objective: $\|y - X\beta\|_2^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2$
- A single blend parameter interpolates between pure ridge and pure lasso

Summary & closing

- Overfitting is a complexity problem: models with too much capacity memorize noise, not signal
- Ridge (L2) and lasso (L1) are the two foundational tools — shrinkage vs. shrinkage + selection
- Both have equivalent readings as constrained optimization and as Bayesian priors
- λ is chosen by cross-validation, not guessed

Next lecture: elastic net and regularization for deep learning, then we put it all into practice in the lab.